General Relativity — Final Examination: Answers

Question 1

Under $x^\mu \to x'^\mu(x)$, the Riemann tensor is a $(1,3)$ tensor, so

$$R'^{\mu}{}_{\nu\rho\sigma} = \frac{\partial x'^\mu}{\partial x^\alpha}\frac{\partial x^\beta}{\partial x'^\nu}\frac{\partial x^\gamma}{\partial x'^\rho}\frac{\partial x^\delta}{\partial x'^\sigma}\, R^{\alpha}{}_{\beta\gamma\delta}.$$

The Ricci tensor in the new coordinates is obtained by setting $\rho = \mu$ and summing:

$$R'_{\nu\sigma} = R'^{\mu}{}_{\nu\mu\sigma} = \left(\frac{\partial x'^\mu}{\partial x^\alpha}\frac{\partial x^\gamma}{\partial x'^\mu}\right)\frac{\partial x^\beta}{\partial x'^\nu}\frac{\partial x^\delta}{\partial x'^\sigma} R^{\alpha}{}_{\beta\gamma\delta}.$$

By the chain rule, $\dfrac{\partial x^\gamma}{\partial x'^\mu}\dfrac{\partial x'^\mu}{\partial x^\alpha} = \dfrac{\partial x^\gamma}{\partial x^\alpha} = \delta^\gamma_\alpha$, so

$$R'_{\nu\sigma} = \frac{\partial x^\beta}{\partial x'^\nu}\frac{\partial x^\delta}{\partial x'^\sigma}\,\delta^\gamma_\alpha R^{\alpha}{}_{\beta\gamma\delta} = \frac{\partial x^\beta}{\partial x'^\nu}\frac{\partial x^\delta}{\partial x'^\sigma}\, R^{\alpha}{}_{\beta\alpha\delta} = \frac{\partial x^\beta}{\partial x'^\nu}\frac{\partial x^\delta}{\partial x'^\sigma}\, R_{\beta\delta},$$

which is exactly the transformation law of a $(0,2)$ tensor.

Question 2

Convention: $\hat\nabla_\mu V^\rho = \partial_\mu V^\rho + \hat\Gamma_{\mu\nu}{}^\rho V^\nu$, $\hat\nabla_\mu \omega_\nu = \partial_\mu\omega_\nu - \hat\Gamma_{\mu\nu}{}^\rho\omega_\rho$. Write $H_{\mu\nu}{}^\rho = g^{\rho\sigma}H_{\mu\nu\sigma}$.

a) Torsion. $T_{\mu\nu}{}^\rho = \hat\Gamma_{\mu\nu}{}^\rho - \hat\Gamma_{\nu\mu}{}^\rho$. The Levi-Civita part is symmetric and cancels, so

$$T_{\mu\nu}{}^\rho = \tfrac{1}{12}g^{\rho\sigma}(H_{\mu\nu\sigma} - H_{\nu\mu\sigma}) = \tfrac{1}{6}H_{\mu\nu}{}^\rho \neq 0$$

(for $H \neq 0$). Hence the connection is *not* torsion-free.

Metric compatibility. Since $\nabla_\mu g_{\nu\lambda} = 0$ for Levi-Civita,

$$\hat\nabla_\mu g_{\nu\lambda} = \nabla_\mu g_{\nu\lambda} - \tfrac{1}{12}H_{\mu\nu}{}^\sigma g_{\sigma\lambda} - \tfrac{1}{12}H_{\mu\lambda}{}^\sigma g_{\nu\sigma} = -\tfrac{1}{12}\left(H_{\mu\nu\lambda} + H_{\mu\lambda\nu}\right) = 0,$$

by antisymmetry of H in its last two indices. So the connection *is* metric compatible.

b) Geodesics. The geodesic equation for $\hat\Gamma$ is

$$\frac{d^2x^\rho}{d\lambda^2} + \hat\Gamma_{\mu\nu}{}^\rho\frac{dx^\mu}{d\lambda}\frac{dx^\nu}{d\lambda} = \frac{d^2x^\rho}{d\lambda^2} + \Gamma_{\mu\nu}{}^\rho\frac{dx^\mu}{d\lambda}\frac{dx^\nu}{d\lambda} + \tfrac{1}{12}g^{\rho\sigma}H_{\mu\nu\sigma}\frac{dx^\mu}{d\lambda}\frac{dx^\nu}{d\lambda} = 0.$$

The last term contracts $H_{\mu\nu\sigma}$, antisymmetric in $\mu\nu$, with $\dot x^\mu\dot x^\nu$, symmetric in $\mu\nu$, so it vanishes: $H_{\mu\nu\sigma}\dot x^\mu\dot x^\nu = -H_{\nu\mu\sigma}\dot x^\mu\dot x^\nu = -H_{\mu\nu\sigma}\dot x^\nu\dot x^\mu \Rightarrow = 0$. Hence the equation reduces to the Levi-Civita geodesic equation; the geodesics coincide. (Only the symmetric part of a connection enters the geodesic equation.)

Question 3

a) The metric components blow up or degenerate at

- $r = 2GM$ (the horizon): $g_{tt} = -f \to 0$, $g_{rr} = 1/f \to \infty$;
- $r = 0$: $g_{tt} \to \infty$, $g_{rr} \to 0$.

(Also $\theta = 0, \pi$, the trivial polar-coordinate singularity of the sphere.) To decide which are physical one computes coordinate-invariant scalars, e.g. the Kretschmann scalar

$$R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} = \frac{48G^2M^2}{r^6}.$$

This is finite ($= 3/(4G^4M^4)$) at $r = 2GM$, so the horizon is only a *coordinate singularity*: nothing local happens to the infalling observer there (for a large black hole tidal forces at the horizon are tiny). It diverges at $r = 0$, which is a genuine *curvature singularity*: tidal forces grow without bound and you would be spaghettified. That is what to worry about—and once inside $r < 2GM$, r is timelike, so reaching $r = 0$ in finite proper time (at most πGM) is unavoidable.

b) Differentiate v:

$$\frac{dv}{dr}\Big|_t = 1 + \frac{2GM}{2GM}\frac{1}{r/2GM - 1} = 1 + \frac{2GM}{r - 2GM} = \frac{r}{r - 2GM} = \frac{1}{f}, \qquad dv = dt + \frac{dr}{f}.$$

So $dt = dv - dr/f$ and

$$-f\,dt^2 = -f\,dv^2 + 2\,dv\,dr - \frac{dr^2}{f}.$$

The last term cancels $+dr^2/f$, giving

$$\boxed{ds^2 = -\left(1 - \frac{2GM}{r}\right)dv^2 + 2\,dv\,dr + r^2(d\theta^2 + \sin^2\theta\,d\phi^2).}$$

At $r = 2GM$ all components are finite and $\det g = -r^4\sin^2\theta \neq 0$, so the metric is perfectly regular there and can be continued smoothly to $0 < r < 2GM$. Ingoing radial null rays are $v = \text{const}$, which cross $r = 2GM$ without incident. This confirms that $r = 2GM$ was an artefact of the (t, r) coordinates, while $r = 0$ remains singular (the metric and curvature invariants still diverge there) in any coordinates.

Question 4

a) The metric has the FLRW form

$$ds^2 = -dt^2 + a(t)^2\left[d\chi^2 + \sin^2\chi\,d\Omega_2^2\right], \qquad a(t) = L\cosh(t/L),$$

with spatial slices round 3-spheres ($k = +1$). Each slice is therefore homogeneous and isotropic, consistent with the cosmological principle (and with observations of the CMB/large-scale structure). Moreover it solves Einstein's equations with a positive cosmological constant $\Lambda = 3/L^2$ (equivalently, the Friedmann equation $\dot{a}^2/a^2 + 1/a^2 = \sinh^2/(L^2\cosh^2) + 1/(L^2\cosh^2) = 1/L^2 = \Lambda/3$). For $t \gg L$, $a \approx \frac{L}{2}e^{t/L}$: accelerated exponential expansion, which describes the late-time dark-energy-dominated universe (and also inflation). The scale factor is $a(t) = L\cosh(t/L)$.

b) Causal structure is determined by light cones, i.e. by which vectors are timelike, null or spacelike. Since $\Omega^2 > 0$,

$$g_{\mu\nu}V^\mu V^\nu = \Omega^2\, \bar{g}_{\mu\nu}V^\mu V^\nu$$

has the same sign as $\bar{g}_{\mu\nu}V^\mu V^\nu$. Hence the same vectors are timelike/null/spacelike in both metrics, the light cones coincide, and the same curves are causal; so the causal relations between points are identical. (Null geodesics are also the same curves up to reparametrisation.)

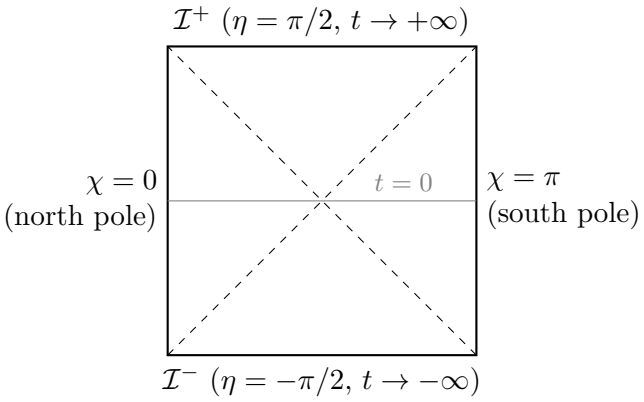
c) With $\cosh(t/L) = 1/\cos\eta$: as t ranges over $(-\infty, \infty)$, $\eta \in (-\pi/2, \pi/2)$ (choosing η with the same sign as t). Also $\sinh(t/L) = \sqrt{\cosh^2 - 1} = \tan\eta$. Differentiating,

$$\frac{1}{L}\sinh(t/L)\, dt = \frac{\sin\eta}{\cos^2\eta} d\eta \;\Rightarrow\; \frac{\tan\eta}{L} dt = \frac{\tan\eta}{\cos\eta} d\eta \;\Rightarrow\; dt = \frac{L\, d\eta}{\cos\eta}.$$

Then

$$ds^2 = \frac{L^2}{\cos^2\eta}\left[-d\eta^2 + d\chi^2 + \sin^2\chi\, d\Omega_2^2\right],$$

conformal (by b)) to the metric of the Einstein static universe $-d\eta^2 + d\chi^2 + \sin^2\chi d\Omega^2$, restricted to $-\pi/2 < \eta < \pi/2$, $0 \le \chi \le \pi$. The Penrose diagram is therefore a square of width π (in χ) and height π (in η); radial null rays are at 45°.



Each interior point represents a 2-sphere; the vertical edges are the poles of the S^3 (not boundaries), and the top and bottom edges $\mathcal{I}^\pm$ are spacelike past and future null infinity. Because $\mathcal{I}^\pm$ are spacelike, there are particle and event horizons: the dashed diagonals are the past/future event horizons of an observer at $\chi = 0$ (resp. $\chi = \pi$). A light ray leaving the north pole at $\eta = -\pi/2$ only just reaches the south pole at $\eta = \pi/2$, so observers at antipodes can never communicate.

Question 5

a) Under $h_{\mu\nu} \to h_{\mu\nu} + \partial_\mu\xi_\nu + \partial_\nu\xi_\mu$, we have $\partial^\mu\bar{h}_{\mu\nu} \to \partial^\mu\bar{h}_{\mu\nu} + \Box\xi_\nu$, so we can always choose ξ with $\Box\xi_\nu = -\partial^\mu\bar{h}_{\mu\nu}$ to reach the Lorenz (de Donder) gauge $\partial^\mu\bar{h}_{\mu\nu} = 0$. Then the second and third terms on the left vanish and

$$\Box\bar{h}_{\mu\nu} = \partial^\rho\partial_\rho\bar{h}_{\mu\nu} = -16\pi G\, T_{\mu\nu}.$$

In vacuum $\Box\bar{h}_{\mu\nu} = (-\partial_t^2 + \nabla^2)\bar{h}_{\mu\nu} = 0$: a wave equation, with solutions $\bar{h}_{\mu\nu} = C_{\mu\nu}e^{ik\cdot x}$, $k^\mu k_\mu = 0$, i.e. ripples in spacetime propagating at the speed of light. Residual gauge freedom ($\Box\xi = 0$) allows one to go to transverse-traceless gauge, $h_{0\mu} = 0$, $h = 0$, $\partial^i h_{ij} = 0$, leaving two physical polarisations $h_+, h_\times$ — gravitational waves.

b) The perturbation is in TT gauge. First, free particles initially at rest stay at fixed coordinate position: the geodesic equation at $U^\mu = (1, 0, 0, 0)$ gives

$$\frac{dU^\mu}{d\tau} = -\Gamma^\mu_{00} = -\tfrac{1}{2}\eta^{\mu\nu}(2\partial_0 h_{0\nu} - \partial_\nu h_{00}) = 0,$$

since $h_{0\nu} = 0$. So the particles remain at $x^i = (0, 0, 0)$, $(L_x, 0, 0)$, $(0, L_y, 0)$. The full metric is

$$ds^2 = -dt^2 + (1 + h_+(t - z))dx^2 + (1 - h_+(t - z))dy^2 + dz^2.$$

Proper distance between first two particles at fixed t, along $y = z = 0$:

$$\Delta s = \int_0^{L_x} \sqrt{g_{xx}}\, dx = \int_0^{L_x} \sqrt{1 + h_+(t)}\, dx = L_x\sqrt{1 + h_+(t)} \approx L_x\left(1 + \tfrac{1}{2}h_+(t)\right),$$

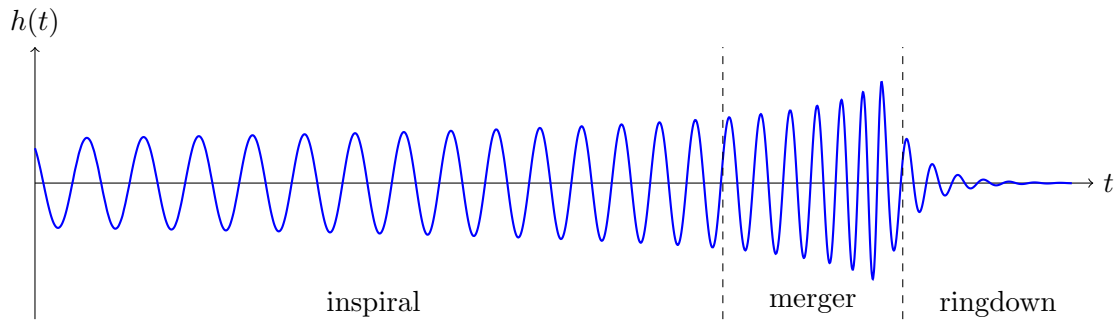
to first order in h_+ (using $z = 0$). Similarly, along $x = z = 0$:

$$\Delta s = \int_0^{L_y} \sqrt{1 - h_+(t)}\, dy = L_y\sqrt{1 - h_+(t)} \approx L_y\left(1 - \tfrac{1}{2}h_+(t)\right).$$

So the + polarisation stretches the x separation while squeezing the y one, and vice versa half a period later.

c) *Physics of the binary.* The two bodies orbit with a time-varying mass quadrupole, so they radiate gravitational waves at twice the orbital frequency, carrying away energy and angular momentum. Losing energy, the orbit shrinks; by Kepler's law the orbital frequency increases, which increases the radiated power (which scales as a high power of the frequency, $P \propto (GM_c\omega)^{10/3}/G$), so the shrinking accelerates. Stages:

- **Inspiral:** slow, adiabatic decay of quasi-circular orbit. The signal is a "chirp": a sinusoid with both frequency and amplitude increasing slowly with time ($f \propto (t_c - t)^{-3/8}$, $h \propto f^{2/3}$). The rate of frequency increase determines the chirp mass $M_c = (m_1 m_2)^{3/5}/(m_1 + m_2)^{1/5}$; the amplitude, combined with this, gives the distance.
- **Merger:** when the bodies reach the innermost stable orbit they plunge together and coalesce (for black holes, horizons merge into a single distorted horizon). Strong-field, highly nonlinear, requires numerical relativity; the amplitude and frequency peak here.
- **Ringdown:** the single remnant (a perturbed Kerr black hole) settles down by emitting quasi-normal modes: a damped sinusoid of roughly constant frequency with exponentially decaying amplitude. Its frequency and damping time determine the final mass and spin.



Detection. LIGO is a Michelson interferometer with arms (4 km) along x and y; the mirrors are suspended so they are effectively freely falling in the horizontal plane, like the test particles of part b). A wave of the form in b) incident along z changes the arm lengths by $\delta L_x = +\frac{1}{2}h_+ L$, $\delta L_y = -\frac{1}{2}h_+ L$, so the round-trip light travel times in the two arms differ, $\Delta T \approx 2h_+ L/c$. The interference of the recombined beams at the photodetector (set near a dark fringe) therefore changes, and the output intensity oscillates in time tracking $h_+(t)$, i.e. the detector measures the strain $h(t) = \Delta L/L \sim 10^{-21}$ directly, producing the chirp waveform sketched above. Matching it against template waveforms reveals the component masses (inspiral), the strong-field dynamics (merger) and the remnant's mass and spin (ringdown).